



Keitaro Hashimoto

Education

- 2020–2023 **Doctor of Engineering (Ph.D.)**, *Tokyo Institute of Technology*, Tokyo, Japan
Post-quantum key exchange protocols for secure messaging. Supervised by Prof. Wakaha Ogata
- 2018–2020 **Master of Engineering**, *Tokyo Institute of Technology*, Tokyo, Japan
Major: Information and communication engineering, specialized in cryptography
- 2014–2018 **Bachelor of Engineering**, *Tokyo Institute of Technology*, Tokyo, Japan
Major: Computer sciences

Experience

- 04/2023–Now **Researcher**, *National Institute of Advanced Industrial Science and Technology (AIST)*, Tokyo, Japan
- 04/2022–03/2023 **JSPS Research Fellowship for Young Scientists**, *Japan Society for the Promotion of Science*, Tokyo, Japan
- 07/2022 **Visiting internship**, *PQShield SAS*, Paris, France
- 06/2020–03/2023 **Research Assistant**, *National Institute of Advanced Industrial Science and Technology (AIST)*, Tokyo, Japan
- 08/2018–09/2018 **Summer internship**, *Nippon Telegraph and Telephone Corporation (NTT)*, Tokyo, Japan
- 08/2017–09/2017 **Summer internship**, *Infosec Corporation*, Tokyo, Japan

Publications

Journals

- [HKKP22] Keitaro Hashimoto, Shuichi Katsumata, Kris Kwiatkowski, and Thomas Prest. An efficient and generic construction for signal's handshake (x3dh): Post-quantum, state leakage secure, and deniable. *Journal of Cryptology*, 35:78 pages, 2022.
- [HO19] Keitaro Hashimoto and Wakaha Ogata. Unrestricted and compact certificateless aggregate signature scheme. *Information Sciences*, 487:97–114, 2019.
- [MKOH25] Koki Matsui, Shoma Kanzaki, Wakaha Ogata, and Keitaro Hashimoto. Round-optimal authenticated key exchange with full forward privacy. *IACR Communications in Cryptology*, 2:28 pages, July 2025.
- [THO24] Kota Takahashi, Keitaro Hashimoto, and Wakaha Ogata. Chosen-ciphertext secure code-based threshold public key encryptions with short ciphertext. *Designs, Codes and Cryptography*, 92:277–301, October 2024.

* 14 April 1995

☎ +81-50-3522-9656 • ✉ keitaro.hashimoto@aist.go.jp
🌐 keitaro-hashimoto.jp • in keitaro-hashimoto • 🌀 kaminomisosiru
ID 0000-0002-2232-9443 • Ⓢ smdloD4AAAAJ • 🐦 kaminomisosiru_

Nationality: Japan

Conference Proceedings

- [AAH⁺25] Kyoichi Asano, Nuttapong Attrapadung, Keisuke Hara, Keitaro Hashimoto, and Yohei Watanabe. Key revocation in registered attribute-based encryption. In *International Conference on Practice and Theory of Public-Key Cryptography (PKC)*. Springer, May 2025.
- [CHH⁺24] Sohto Chiku, Keisuke Hara, Keitaro Hashimoto, Toi Tomita, and Junji Shikata. How to apply fujisaki-okamoto transformation to registration-based encryption. In Markulf Kohlweiss, Roberto Di Pietro, and Alastair Beresford, editors, *International Conference on Cryptology And Network Security (CANS)*, pages 145–165. Springer, September 2024.
- [FHHY25] Yuuki Fujita, Keisuke Hara, Keitaro Hashimoto, and Kyosuke Yamashita. Multi-designated verifier ring signature: Generic construction from standard primitives. In *International Workshop on Security (IWSEC 2025)*. Springer, November 2025.
- [GTH26] Milan Gonzalez-Thauvin and Keitaro Hashimoto. Asymmetric message franking in the plain model: Generic and efficient constructions. In *Annual International Cryptology Conference (Crypto)*, August 2026.
- [HHM⁺26] Keisuke Hara, Keitaro Hashimoto, Takahiro Matsuda, Wataru Nakamura, and Kenta Takahashi. (re-)formalization and construction of reusable and robust threshold fuzzy extractors. In *Applied Cryptography and Network Security (ACNS) Conference*, June 2026.
- [HKKP21] Keitaro Hashimoto, Shuichi Katsumata, Kris Kwiatkowski, and Thomas Prest. An efficient and generic construction for signal's handshake (x3dh): Post-quantum, state leakage secure, and deniable. In Juan A. Garay, editor, *International Conference on Practice and Theory of Public-Key Cryptography (PKC)*, pages 410–440. Springer, May 2021.
- [HKNW26] Keitaro Hashimoto, Shuichi Katsumata, Guilhem Niot, and Thom Wiggers. Revisiting pq wireguard: A comprehensive security analysis with a new design using reinforced kems. In *IEEE Symposium on Security and Privacy (IEEE S&P 2026)*. IEEE, May 2026.
- [HKP⁺21] Keitaro Hashimoto, Shuichi Katsumata, Eamonn W. Postlethwaite, Thomas Prest, and Bas Westerbaan. A concrete treatment of efficient continuous group key agreement via multi-recipient pkes. In *The ACM Conference on Computer and Communications Security (ACM CCS)*. ACM DL, November 2021.
- [HKP22] Keitaro Hashimoto, Shuichi Katsumata, and Thomas Prest. How to hide metadata in mls-like secure group messaging: Simple, modular, and post-quantum. In *The ACM Conference on Computer and Communications Security (ACM CCS)*. ACM DL, November 2022.
- [HKPP25] Keitaro Hashimoto, Shuichi Katsumata, and Guillermo Pascual-Perez. Exploring how to authenticate application messages in mls: More efficient, post-quantum, and anonymous blocklistable. In *Usenix Security 2025*, Berkeley, California, USA, August 2025. USENIX.
- [HKW25] Keitaro Hashimoto, Shuichi Katsumata, and Thom Wiggers. Bundled authenticated key exchange: A concrete treatment of (post-quantum) signal's handshake protocol. In *Usenix Security 2025*, pages pp.1–54, Berkeley, California, USA, August 2025. USENIX.
- [HYH25] Keitaro Hashimoto, Kyosuke Yamashita, and Keisuke Hara. Foundations of multi-designated verifier signature: Comprehensive formalization and new constructions in subset simulation. In *IEEE Computer Security Foundations Symposium (CSF)*. IEEE, June 2025.

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🆔 0000-0002-2232-9443 • Ⓢ smdloD4AAAAJ • 🐦 kaminomisosiru_

Nationality: Japan

- [ZHO24] Xichen Zhang, Keitaro Hashimoto, and Wakaha Ogata. Security model for authenticated key exchange, reconsidered. In Clemente Galdi and Duong Hieu Phan, editors, *International Conference on Security and Cryptography for Networks (SCN)*, pages 251–271. Springer, September 2024.

Talks

My Talks in Peer-Reviewed Conferences

- 06/2026 **ACNS**, *(Re-)formalization and construction of reusable and robust threshold fuzzy extractor*, New York, USA
- 05/2026 **IEEE S&P**, *Revisiting PQ Wireguard: A Comprehensive Security Analysis with A New Design Using Reinforced KEMs*, San Francisco, USA
- 08/2025 **Usenix Security**, *Exploring how to authenticate application messages in mls: More efficient, post-quantum, and anonymous blocklistable*, Seattle, USA
- 06/2025 **IEEE CSF**, *Foundations of multi-designated verifier signature: Comprehensive formalization and new constructions in subset simulation*, Santa Cruz, USA
- 09/2024 **SCN**, *Security Model for Authenticated Key Exchange, Reconsidered.*, Amalfi, Italy
- 08/2024 **CFail**, *Towards a Tightly Secure Signature in Multi-User Setting with Corruptions Based on Search Assumptions*, Saint Barbara, USA
- 11/2022 **ACM CCS**, *How to Hide MetaData in MLS-Like Secure Group Messaging: Simple, Modular, and Post-Quantum*, Los Angeles, USA
- 11/2021 **ACM CCS**, *A Concrete Treatment of Efficient Continuous Group Key Agreement via Multi-Recipient PKEs*, Virtual
- 05/2021 **PKC**, *An Efficient and Generic Construction for Signal's Handshake (X3DH): Post-Quantum, State Leakage Secure, and Deniable*, Virtual

Invited Talks

- 05/2026 **SKECH**, *Computational Security Models for PQ WireGuard*, Bertinoro, Italy
- 01/2026 **New Concepts and Paradigms of Security for Next Generation**, *Towards Social Implementation of Secure IoT Devices*, Tokyo, Japan
- 03/2025 **IETF**, *Authentication in MLS and Its Variants.*, Bangkok, Thai
- 09/2023 **Forum on Information Technology (FIT)**, *How to Hide MetaData in MLS-Like Secure Group Messaging: Simple, Modular, and Post-Quantum*, Virtual
- 09/2022 **Workshop on Cryptography and Information Security (WCIS)**, *A Concrete Treatment of Efficient Continuous Group Key Agreement via Multi-Recipient PKEs*, Virtual
- 07/2022 **ENS de Lyon**, *A Concrete Treatment of Efficient Continuous Group Key Agreement via Multi-Recipient PKEs*, Lyon, France
- 09/2021 **SCIS/CSS Invited Session in IWSEC**, *Design and Implementation of a Post-Quantum Authenticated Key Exchange Protocol for Signal*, Virtual

Peer-Reviewed Conference Talks Presented by My Co-authors

- 05/2026 **PrivCrypt**, *Efficient Asymmetric Message Franking in the Plain Model*, Rome, Italy, Presented by Milan Gonzalez-Thauvin
- 09/2025 **NIST PQC Standardization Conference**, *A Comprehensive Study of the Signal Handshake Protocol: Bundled Authenticated Key Exchange*, Virtual, Presented by Ida Tucker
- 03/2024 **RWC**, *More efficient protocols for post-quantum secure messaging*, Toronto, Canada, Presented by Thomas Prest

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🌐 keitaro-hashimoto.jp • in keitaro-hashimoto • 🎧 kaminomisosiru
🆔 0000-0002-2232-9443 • 📄 smdloD4AAAAJ • 🐦 kaminomisosiru_

Nationality: Japan

- 03/2023 **RWC**, *Metadata Protection for MLS and Its Variants*, Tokyo, Japan, Presented by Shuichi Katsumata
- 02/2023 **ETSI/IQC Quantum Safe Cryptography Event**, *A Post-Quantum Construction for Signal's Handshake (X3DH)*, Virtual, Presented by Thomas Prest
- 12/2022 **NIST PQC Standardization Conference**, *An Efficient and Generic Construction for Signal's Handshake (X3DH)*, Virtual, Presented by Thomas Prest

Honors and Awards

- 2026 Outstanding Paper Award in Department of Information Technology and Human Factors, AIST
- 2025 Special prize of Tsujii Shigeo Security Award 2025
- 2024 Best Paper Award in Computer Security Symposium 2024 (CSS2024)
- 2024 Seiichi Tejima Research Award Doctoral Dissertation Award
- 2023 Grand prize of Tsujii Shigeo Security Award 2023
- 2022 Tsujii Shigeo Security Award 2022
- 2021 Tsujii Shigeo Security Award 2021
- 2021 Best Paper Award in Symposium on Cryptography and Information Security 2021 (SCIS 2021)
- 2020 Outstanding Student Award at Tokyo Institute of Technology

Patents

- US Patent App. 19/033,305 K. Hashimoto, S. Katsumata, T. Prest: Method for hiding the identity of communication devices in a continuous group key agreement

Internship Supervision

- 03/2025–05/2025 **Milan Gonzalez-Thauvin**, *ENS de Lyon*, Lyon, France
Topic: Anonymous signatures with application to secure messaging

Academic Services

- Journal Reviewer Designs, Codes and Cryptography, ACM Transactions on Privacy and Security, IEICE TRANSACTIONS on Fundamentals of Electronics, Communications and Computer Sciences, IPSJ Journal
- Conference Reviewer CRYPTO, EUROCRYPT, ASIACRYPT, PKC, CANS, ACNS

Languages

- Japanese Native
- English Intermediate
- French Beginner
- Taiwanese Beginner
- Mandarin

Certifications

- 03/2021 **Improve Your English Communication Skills Specialization**, Coursera, A3ZGXJ8RWW5T

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 ID 0000-0002-2232-9443 • Ⓜ️ smdloD4AAAAJ • 🐦 kaminomisosiru_

Nationality: Japan

- 03/2021 **Introduction to Mathematical Thinking**, *Coursera*, WQY3UEVLZSEE
12/2015 **Applied Information Technology Engineer**, *Ministry of Economy, Trade and Industry*,
AP-2015-10-03112
10/2014 **Fundamental Information Technology Engineer**, *Ministry of Economy, Trade and
Industry*, FE-2014-10-04834

Computer Skills

Programming Java, Rust, Python

Typesetting $\LaTeX$ / $\TeX$

References

- Wakaha Ogata (Ph.D. advisor): ogata.w.aa@m.titech.ac.jp

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☎ +81-50-3522-9656 • ✉ keitaro.hashimoto@aist.go.jp
🌐 keitaro-hashimoto.jp • in [keitaro-hashimoto](https://www.linkedin.com/company/keitaro-hashimoto) • 🌀 [kaminomisosiru](https://kaminomisosiru.com)
id 0000-0002-2232-9443 • Ⓜ [smdloD4AAAAJ](https://github.com/smdloD4AAAAJ) • 🐦 [kaminomisosiru_](https://twitter.com/kaminomisosiru_)
Nationality: Japan